NALLA NARASIMHA REDDY
Education Society's Group of Institutions - Integrated Campus
(UGC AUTONOMOUS INSTITUTION)

FINAL REVIEW

Batch No : B20

S.K Irfan Baba (227Z1A66B5)
Siri Vennela (227Z1A66C6)
Sarva Darshan (227Z1A66B4)

Under the Guidance of :-
Mrs. G. Manasa

System Overview

- **Purpose:** Enable early detection of Autism Spectrum Disorder (ASD) in toddlers through a simple online screening tool.
- **Process Flow:** User Data → Preprocessing → ML Prediction (XGBoost) → ASD Risk Level.
- **Frontend:** Developed using Flask with HTML/CSS to create a clean, responsive interface.
- **Backend:** Processes input and interacts with the trained XGBoost model for real-time predictions.
- **Output:** Displays ASD risk level (e.g., mild, moderate, severe) along with recommendations or next steps.

Testing Done

- **Unit Testing:** Tested individual functions such as ***predict_autism()*** to ensure accurate logic and output.
- **Integration Testing:** Verified smooth data flow and interaction between **frontend and backend components.**
- **System Testing:** Conducted end-to-end testing of the **entire application** workflow to ensure reliability.
- **User Acceptance Testing (UAT):** Collected feedback from **caregivers** and **educators** to assess usability and clarity of results.

Test Cases

Test Case ID	Description	Input	Expected Output	Result
TC01	Valid Data	Complete and correctly formatted toddler details	Risk Level (Low / Medium / High)	Pass
TC02	Empty Field	Missing value in age field	Error Message prompting user to complete input	Pass
TC03	Invalid Format	Text entered in a numeric field (e.g., "Two" instead of 2)	Validation Error or form highlight	Pass
TC04	ML Accuracy	Pre-labeled test dataset entries	Model predicts correct labels with high accuracy	Pass
TC05	Load Test	50+ users submitting data simultaneously	Consistent server performance and response time	Pass
TC06	Model Availability	API call to prediction route while model is loading	Graceful fallback or "Model Loading" message	Pass
TC07	UI Feedback	After form submission	Result display with prediction and risk factors	Pass

Test Cases

Test Case ID	Description	Input	Expected Output	Result
TC08	Risk Factor Breakdown	Valid input with mix of 0s and 1s for Q-Chat	List of at-risk questions and total score	Pass
TC09	Persistence Check	Predict, reload page, predict again	No crash or memory issue; consistent output	Pass
TC10	Invalid Route	GET on Predict data	Error on Delay	Pass
TC11	Security Test	Script injection	No execution	Pass
TC12	API Timeout	Slow Network	Fallback or fast	Fail
TC13	Incomplete form	Skip questions	Error Message	Fail
TC14	Corrupt Model	Bad model file	Error Message	Fail

Results and Output

- **Accurate prediction** of Autism Spectrum Disorder (ASD) risk categorized as **Low**, **Medium**, or **High**.
- Provides **fast and interpretable feedback** based on user responses.
- Fully **responsive design** ensures smooth experience on both **mobile and desktop devices.**
- Serves as an **accessible and practical screening tool**, offering early guidance for parents and caregivers.

Future Enhancements

- **Incorporate speech** and **facial expression analysis** for more comprehensive autism detection.
- Extend support to a **broader age range**, including older children and adolescents.
- **Develop a mobile app** version for wider accessibility and offline screening.
- Integrate with pediatric **Electronic Health Records (EHRs)** for seamless clinical use.
- Add an **AI-powered chatbot** to assist users with queries and guide them through the screening process.

References

- UCI ASD Dataset: Source of real-world data for autism screening model training.

Q-CHAT-10 Research Papers:

- Allison, C., et al. (2012). *The Quantitative Checklist for Autism in Toddlers (Q-CHAT)*. Psychological Assessment.
- Baron-Cohen, S., et al. (2001). *The Checklist for Autism in Toddlers (CHAT)*. Journal of Autism and Developmental Disorders.

AI in Autism Research:

- Kosmicki, J. A., et al. (2015). *Searching for a minimal set of behaviors for autism detection through feature selection-based machine learning.* Translational Psychiatry.

References

Machine Learning for Autism Detection:

- Thabtah, F. (2017). *Autism spectrum disorder screening: Machine learning adaptation and DSM-5 fulfillment*. Proceedings of the 1st International Conference on Medical and Health Informatics.
- Bone, D., et al. (2016). *Use of machine learning to improve autism screening and diagnostic tools: Effectiveness, challenges and future directions.* Journal of Autism and Developmental Disorders.

Conclusion

- Successfully **implemented an end-to-end ASD screening system** combining machine learning and web technologies.
- Showcased the **synergy between ML and web development** to create a functional, real-time application.
- Designed to **empower caregivers and parents** by providing early insights into ASD risk.
- Demonstrated the **complete Software Development Life Cycle (SDLC)** — from problem identification and planning to implementation, testing, and deployment.